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Designing the public sector to promote sustainability transitions: Institutional principles and a case study of ARPA-E

Brendan Haley[a,b]

[a] Smart Prosperity Institute, Institute of Environment, University of Ottawa, 1 Stewart Street, Ottawa, Ontario, K1N 7M9, Canada
[b] School for Resource and Environmental Studies, Dalhousie University, Kenneth C. Rowe Management Building, 6100 University Ave, PO BOX 15000, Halifax, Nova Scotia, B3H 4R2, Canada

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ABSTRACT

Sustainability transition perspectives justify policies tailored to technological innovation systems and socio-technical niches and regimes. These "technology-specific" policies place high demands on the capabilities of the state. This paper examines how public sector institutions can be designed and operated to avoid potential pitfalls and effectively implement technology-specific policies. The paper presents a list of ten institutional design principles and their relation to sustainability transition studies. How the principles are achieved in practice and how they interact within a governance system is then explored through a case study of the Advanced Research Projects Agency-Energy (ARPA-E) in the United States. The case demonstrates that an organization's formal and informal rules can make the design principles mutually reinforcing, while a stronger political consensus is needed to manage tensions between accountability and stability. The institutional design principles can inform future case studies exploring how the state implements technology specific policy in different contexts.

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1. Introduction

Sustainability transitions perspectives emphasize the need to tailor policies toward the particular challenges found within technological innovation systems (Carlsson and Stankiewicz, 1991; Bergek et al., 2008) or socio-technical regimes and niches (Geels, 2002; Smith and Raven, 2012). These perspectives emphasize the role of policies that are targeted, or "technology-specific," in contrast to those that seek to meet the difficult standard of "technology neutrality" (see Azar and Sandén, 2011; Jacobsson and Bergek, 2011).

However, technology-specific policies place high demands on the state's capability to continuously monitor innovation systems and undertake targeted policy actions. Even if more activist governments following technology-specific policy approaches can be theoretically justified, others warn that such policies can be derailed by "government failures" (Krueger, 1990; Tullock et al., 2002). Advocates of government-directed technological change respond to these concerns by pointing toward "organizational innovation" (Chang, 1994). Given this debate, it is important that sustainability transitions scholars not only outline the theoretical arguments for technology-specific policies, but also explain how such policies can be effectively *implemented*. Indeed, Jacobsson and Bergek (2011) highlight researching the "competence, organisation and integrity

E-mail address: bhaley@dal.ca

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of public policy bodies" as a key sustainability transitions research priority. This research requires an understanding of the pitfalls that a technology-specific agenda could encounter during implementation, and how institutional design could help avoid these pitfalls.

This paper's first purpose is to propose ten institutional design principles for the effective implementation of technology-specific policies for sustainability transitions. Its second purpose is to demonstrate how these principles are achieved in practice and how they interact through a case study of the Advanced Research Project Agency-Energy (ARPA-E) in the United States. Section 2 briefly reviews arguments on government involvement in technological and industrial development, and asserts that these perspectives reveal the need for a stronger focus on institutional design and democratic decision-making processes, rather than a retreat from state involvement in shaping sustainable technology futures. Section 3 outlines the methodological approach, and Section 4 proposes ten public sector institutional design principles and their relevance for sustainability transitions policy. Section 5 presents a case study of ARPA-E, which demonstrates how these seemingly contradictory principles can co-exist, producing balance and synergy. The case also demonstrates how tensions between principles create management challenges. The paper concludes with lessons from the case study and ideas for future research.

2. Technology-specific policies: their rationale and potential pitfalls

The urgent need to transition toward a low-carbon economy has reintroduced the importance of technology-specific policy approaches. These approaches contrast with "technology-neutral" policies that seek to avoid "picking winners" by providing broad signals through mechanisms such as carbon pricing and generalized support for research and development. Sandén and Azar (2005) argue that technology-neutral policies promote the adoption of near-commercial technologies, but fail to produce new technological options. Jacobsson and Bergek (2011) note that sustainability transitions require the development of a large suite of technologies in a limited period of time. They call for "multifaceted and specific" policy interventions because the innovation systems associated with each technology will face different barriers and vary in their development stages.[2] The technological innovation systems (TIS) approach provides a tool to diagnose system weaknesses that require policy attention (Bergek et al., 2008). The multi-level perspective (MLP) on sustainability transitions (Geels, 2011, 2002) also emphasizes the importance of social and technological learning within niche protected spaces (Smith and Raven, 2012), and assesses how these niches evolve to challenge the dominant technological paradigm or "regime." Yet, neither approach offers guidance on how the state can effectively implement the recommended policy actions.

Technology-specific policies require an active role for the state in supporting innovation and guiding economic transformations. The state has played a leading role in technological development throughout history (Block and Keller, 2011; Mazzucato, 2013; O'Riain, 2004; Perez, 2002). However, government attempts to promote technological innovation have also resulted in spectacular failures (Lerner, 2009; Lipsey and Carlaw, 1996). This section will discuss four categories of potential technology-specific policy pitfalls: information failures, political capture, principal-agent problems, and challenges associated with risk aversion stemming from policy ambiguity and complexity. These pitfalls present useful warnings about what could go wrong, and recognition of these problems rules out simplistic views of the state as benevolent or all-knowing. However, this section also argues that these warnings fail to justify avoidance of technology-specific policies. Rather, acknowledging these pitfalls calls for a more careful consideration of how public institutions are designed and operated.

The state's inability to acquire sufficient information to make informed decisions is often presented as an argument against activist government policies. For instance, the Austrian school of economics pointed to information limitations to explain the infeasibility of central planning, and highlighted the role of prices in transmitting information (Hodgson, 1998; Lavoie, 1985). Yet, incomplete information is pervasive across both public and private sectors. Private firms have information deficits, and while market prices transmit valuable information, non-market institutions are critically important for developing new products and processes (Lundvall, 1992). In practice, both firms and governments manage technological change by creating associative structures that enable learning and information exchange (Meadowcroft, 1997; Cooke and Morgan, 1998; Rycroft and Kash, 1999). Removing government from such networks could restrict information flow to both sectors.

Political capture is a second pitfall that occurs when interest groups influence governments to take advantage of policy-induced rents (Buchanan et al., 1980). The rents created by the state entice entrepreneurs to engage in lobbying, and even corruption and bribery, rather than productivity enhancing activities (see Krueger, 1974). The concern over political capture suggests that politics is ubiquitous, yet resorting to "technology-neutral" policies is unlikely to prevent certain groups from gaining undue influence over the state. Azar and Sandén (2011) argue that policies seldom meet the standard of technology-neutrality, and that less targeted policies favour incumbent technologies. The notion of the regime (Kemp, 1994) in sustainability transition studies highlights the lock-in effects of dominant technologies and incumbent players. Non-action by government reinforces status-quo regimes, with the political implication of supporting the interests associated with existing technologies.

Skeptics of government intervention also highlight the prospect for uncontrollable, expanding bureaucracies with little accountability to elected representatives. Tullock (1965) discusses information problems within the state, whereby direc-

[2] See also Acemoglu et al. (2012), who call for a climate policy mix that includes both carbon taxes and more specific measures to influence the direction of technical development.

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tions from elected representatives and bureaucratic superiors are lost or slanted, and where civil servants have an interest in selectively filtering information presented to superiors. Niskanen (1974) presents a model of bureaucrats increasing their individual self-interest by maximizing their agencies' budgets. These theories highlight a third pitfall that can be categorized as principal-agent problems. These problems can occur because bureaucratic agents have information and implementation advantages over political principals. Yet there is little empirical evidence that civil servants use these advantages to avoid political direction (Wood and Waterman, 1994), and evidence of the budget-maximizing bureaucrat is mixed (Blais and Dion, 1990; Dolan, 2002; Ingvar Jacobsen, 2006). The bureaucracy's informational advantages can be a powerful aid to democratic decision making, since political parties are unlikely to have adequate resources to fully interrogate policy issues.[3] Indeed, it is the public service's role to provide "fearless advice" (Himelfarb, 2002; Savoie, 2015) so political leaders can make informed policy decisions.

The potential for principal-agent problems demands a closer look at the role of bureaucracies in policymaking and the motivations of civil servants. Civil servants do not simply implement the orders of political principles; they play a key role as democratic agents who shape policy through design and implementation (Waldo, 1946 Wilson, 1989). Thus, civil servants' motivations and integrity are important. While some civil servants are surely motivated by personal interests, they may also be motivated by professionalism, idealism and altruism, as well as the desire for pride and prestige (McGregor, 1960; Schmitz et al., 2015). The model of the selfish bureaucrat is dangerous, as it can become self-fulfilling by decreasing the public service's legitimacy (Haque, 1996). The selfish bureaucrat model can also justify policies, like increased monitoring, that crowd out unselfish motivations (see Chang, 2002).

Principal-agent models also poorly represent policymaking complexity, which presents a fourth policy pitfall. Bureaucrats face implementation challenges because they work in policy arenas with significant ambiguity (Zahariadis, 2014). Ambiguity must be expected because the state accommodates elected representatives and departments with a variety of different political perspectives, problem definitions, and preferred solutions. Rittel and Webber (1973) described how state planners must deal with "wicked problems" that are inherently different from problems confronted by scientists and engineers. In the policy realm, plural interests define the public good and societal problems in different ways. Policy planners must continuously search for "good" solutions (rather than "optimal" or final solutions) to ill-defined problems. In complex policy environments decisions often contradict each other and vacillate, as they represent the outcomes of ongoing political negotiations between several areas of government representing different interests.[4] Civil servants can react to ambiguity by resorting to familiar and established procedures, even if these procedures negatively impact the effectiveness of policy implementation (Atkinson and Powers, 1987). The development of a "passive risk-avoidance approach" (Savoie, 2015) can permeate public organizations, especially if the primary concern of senior managers and politicians is avoiding political surprises and blame (Hood, 2010). Thus, the bureaucratic reaction to a complex and ambiguous policy environment coupled with a blame-generating political environment can prevent the development of an activist, forward-looking state.

Yet, the state is the actor in society most capable of grappling with complex "wicked" problems[5] because the public sector is better equipped than non-governmental sectors that lack the resources or motivation to work on these intractable societal puzzles (Savoie, 2015). Technology-specific policy frameworks are designed to grapple with complexity through monitoring, experimentation, diagnoses of context-specific barriers, and concern for political legitimacy. The potential pitfalls associated with policy complexity suggest that implementing technology specific policies requires an institutional environment that rewards appropriate risk-taking, while ensuring democratic accountability and representation.

3. Method

This paper explores how the four pitfalls discussed above can be avoided by drawing on insights from existing literature and a case study. The paper proposes ten institutional design principles, developed from a review of academic and grey literatures, with specific reference to sustainability transition studies. Keyword searches in topic areas; like industrial policy or the developmental state; helped identify sources; but were of limited value because institutional design for innovation and technology policy has not yet coalesced into a distinct field of study with consistent language. The intellectual community associated with this topic is scattered across several disciplines; including public administration; political economy; industrial policy; developmental economics; sustainability transitions and innovation studies. Relevant sources for the literature review were identified from the author's knowledge and by soliciting suggestions from experts. References and citations in core works; together with targeted keyword searches; yielded further materials. The key principles explicitly or implicitly listed in the sources consulted were separated and consolidated into a manageable list.

Following discussion of these design principles is a case study of the US Department of Energy agency ARPA-E. The author collected case information from primary materials, including reports, testimony and presentations, and secondary materials available in academic publications, newspapers, industry magazines, and public policy reports. The principles are used as a framework to explore the ARPA-E case, and should not be viewed as independent variables capable of explaining

[3] Weber (1978) espoused the technical advantages of bureaucracy's "knowledge growing out of experience in the service" (p. 225), yet recognized the need for bureaucracy's power to be checked by political representatives and entrepreneurs.
[4] See the bureaucratic politics model pioneered by Allison (1971).
[5] See Levin et al. (2012) for a discussion of climate change as a "super wicked problem".

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organizational performance. As presented by Flyvbjerg (2001), case studies provide examples that facilitate learning about context-specific situations where judgment must be exercised. Thus, the purpose of this case study is to promote learning about the messy, context-dependent process of designing and operating public sector institutions to promote sustainability transitions. ARPA-E was selected because it illustrates how institutional design works in practice, and this example has the potential to act as a key reference for future studies. The United States national context also facilitates the exploration of tensions. The US is an industrialized, democratic country without an ideological predisposition towards "industrial policy" or "developmental state" based policies, associated with technology specific approaches.[6] Thus, ARPA-E must contend with the criticisms and fears associated with government technology involvement discussed in the earlier section.

4. Institutional design principles

Before presenting the institutional design principles, it is worthwhile to understand what is meant by "institution." Institutions are a broader concept than organizations. The concept describes the formal and informal rules that govern society (Hodgson, 2004; North, 1990). This includes rules created by the public sector, or simply the norms that govern how things are done. While institutions and organizations are not the same thing, many public sector organizations are also institutions because they embody collective decisions and values (Frederickson et al., 2016, pp. 72–74). In this paper, the concept of institutions helps explore the structure of public sector organizations and their internal functioning, as well as how these organizations relate to non-governmental actors.

Ten institutional design principles will be discussed:

1) Comprehensiveness
2) Flexibility
3) Autonomy from Short-term Political Pressure
4) Mission Orientation
5) Embeddedness within Policy Networks
6) Autonomy from Private Interests
7) Competence
8) Credibility
9) Stability
10) Accountability

The sub-sections below describe each principle, and discuss how they help alleviate the pitfalls discussed in Section 2 as well as how they relate to sustainability transitions studies.[7] Relevant case examples are provided, to describe the principles.

4.1. Comprehensiveness

Technology-specific policies require the state to have a comprehensive outlook and the ability to wield a complete set of policy tools. The systems theories used in sustainability transitions studies highlight the importance of considering all factors that influence technological development, to avoid information problems and critical gaps. Thus, the technological innovation systems approach analyzes various social and technological processes, and counsels that the state develop a comprehensive "policy mix" rather than relying on one or two instruments (see Kivimaa and Kern, 2016). Comprehensiveness is also needed in an environment with ambiguous policy objectives and wicked problems, where the consequences of policy actions are uncertain. To manage complexity, the sustainability transitions perspective emphasizes comprehensiveness with respect to the diversity of socio-technical options explored and the variety of actors involved in "reframing problems and solutions" (Loorbach, 2010, p. 168).

Comprehensiveness presents a challenge for policy implementation and public administration. A government unit might be particularly adept at implementing a policy instrument, or specialized in a stage of innovation. However, as stated by Jacobsson and Bergek (2011, p. 54), "no ministry or other government body can be expected to have all the required instruments in its arsenal."

Achieving comprehensiveness might require centralization or coordination across various government entities. There are examples of centralized "pilot agencies" promoting economic transitions (Chang, 2010). However, Breznitz and Orston (2013) and Block (2008) argue that pilot agencies helped countries catch-up within existing industries, yet could work *against* the type of information flow and experimentation required for radical innovation and discovery of new technological trajectories.[8] Centralization could increase potential for capture by vested interests and prevent novel partnerships between public and private actors. Block (2008) and O'Riain (2004) discuss a "networked" state with a decentralized structure of labs,

[6] However, Block (2008) presents evidence that industrial policy is widely practiced in the US.
[7] This list of principles is discussed in the Canadian context in Haley (2016)
[8] See Wong (2011) for discussion of the struggles of traditional developmental states in biotechnology.

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agencies, and departments. Block (2008, p. 174) notes that the redundancy that accompanies decentralization is beneficial because it increases the potential that an idea may flourish despite lock-out by some government units.

Foray et al. (2012, p. 1701) call for combining a centralized administrative structure, setting broad priorities and monitoring progress, with sufficient decentralization to support a diverse range of technologies and actors. Achieving the right balance is challenging. Decentralization could see sub-units with different problem definitions and preferred solutions contradicting high-level policy objectives. Decentralization increases the likelihood that an idea will gain initial support, but could result in hand-off problems when other departments need to be involved.[9] Kern and Howlett's (2009) review of the Netherlands found that the energy transition policy agenda was jumbled with other contradictory and incoherent policy objectives, and that being both comprehensive and coherent involves "tough political struggles" (p. 404) over policy goals and instruments.

4.2. Flexibility

Flexibility is needed to respond to new dynamics in innovation systems and to terminate unpromising projects. Information is imperfect, creating the need for policy learning. This is especially true in wicked problem environments, where the end results of policy actions cannot be easily predicted.

Evolutionary perspectives within sustainability transitions frameworks suggest that policymakers must be flexible. The multi-level perspective emphasizes time-specific "windows of opportunity," when dominant socio-technical regimes are open to change (Geels, 2002). Policymakers must be prepared to exploit these windows by scaling up niche-level technologies. The TIS perspective emphasizes that innovation weaknesses change over time (Hekkert et al., 2011), requiring different types of policy intervention.

Flexibility is also important to cancel projects that show little promise, to allay concerns of runaway bureaucracy per the principal-agent models previously discussed. Lipsey and Carlaw (1996) present cases of projects that continued to receive support because ambiguous policy objectives made cancelling them difficult. For instance, the British Atomic Energy Authority's objective to develop a gas-cooled nuclear reactor mixed technological development objectives with political prestige and military rationales. Multiple objectives, combined with top-down political direction, made it difficult for civil servants to respond to cost overruns and environmental questions.[10] Recognizing that failure is inevitable when exploring uncertain technologies, Lipsey and Carlaw (1998) suggest that policymakers celebrate situations where failure is identified in a timely fashion and support for the failing project is terminated.

Giving civil servants greater discretion to quickly cut-off or scale-up projects can achieve flexibility; yet this can challenge traditional notions of public administration, which call for hierarchical control and impersonal rules to prevent favouritism and corruption (Weber, 1978). Thus, flexibility must be complemented by other institutional design principles.

4.3. Autonomy from short-term political pressure

Public sector decision making is a political endeavour. Yet a policy program can encounter different types of political pressures, depending on institutional location and organizational structure (see Seidman, 1998). As civil servants travel up the government hierarchy, they must become adept at managing controversies and conflicting objectives (Manzie and Hartley, 2013; Savoie, 2015). As discussed in the last section, there is a danger that too much political control could make prestige an objective, rather than an outcome of technology-specific policies, and that this could decrease the flexibility needed to cancel non-performing projects (Lipsey and Carlaw, 1996). There is also a danger of excessive aversion to technological failures, even though they are normal occurrences during innovation searches (see Rodrik, 2014). Senior civil servants and political representatives will seek to avoid blame, while civil servants tasked with implementation might react to political environments and unclear objectives by substituting safe protocols for potentially transformative initiatives.

One way to address a complex policy environment is to reduce the complexity through administrative arrangements that provide some independence from short-term political pressure. Arm's-length sustainability organizations include the UK Carbon Trust, created as a not-for-dividend private company, and Sustainable Development Technology Canada, a separate foundation. Breznitz and Ornston's (2013) review of innovation agencies in Israel and Finland emphasized that they were successful because of their position on the "periphery" of government administration. This position created room for flexibility and experimentation, and created a "low profile" away from short-term political controversies and attention by powerful interests. Examples exist of organizations without arm's-length organizational structures that have nevertheless operated in institutional environments with strong political autonomy. Japan's Ministry of International Trade and Industry (MITI) was empowered by a tradition of bureaucratic dominance over politics (Johnson, 1982), and Rodrik (2014) argues that the US Defense Advanced Research Projects Agency maintains its independence because of its high level of professionalism and expertise.

A *niche* space within the public sector can create autonomy from short-term political pressures. Transitions frameworks have emphasized the importance of technological and social niches (Kemp et al., 1998), yet have not discussed how to create

[9] Breznitz and Ornston (2016) discuss "irrelevant innovator" agencies, which produce new ideas but lack implementation capabilities.
[10] In this example, political principals, rather than bureaucratic agents, increased government expenditure.

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similar niches within the state. The Netherlands policy agenda inspired by transitions frameworks began as a "policy niche," but was not implemented through the creation of a separate organizational entity. The powerful Ministry of Economic Affairs assumed the role of transitions manager, followed by an Interdepartmental Program Directorate to coordinate activities across six different ministries (van der Loo and Loorbach, 2012).

While niche organizations within the state might more easily align with socio-technical niches, such organizations could lack the resources and policy authority required to scale-up promising environmental innovations. Breznitz and Orston (2016) argue that in an increasingly politicized "innovation policy" environment, agencies must choose between the independence to experiment and access to the resources needed to implement policy ideas. Thus, autonomy from short-term political interests might conflict with the comprehensiveness principle previously discussed.

4.4. Mission orientation

Faced with policy ambiguity and unclear objectives, public servants can prioritize safe procedures over effective implementation strategies. Wilson's (1989) review of successful and unsuccessful public organizations concluded that effective managers were able to define a clear mission and provide implementers with the autonomy needed to deliver on the outlined objective. Mazzucato's (2013,2015) vision of the "entrepreneurial state" calls for creating "mission oriented" organizations that clearly understand their role within innovation systems and have the capacity to "think big."

A mission orientation can avoid the self-serving bureaucrat model by appealing to other motivators. Narayanamurti et al. (2009) argue that a mobilizing mission acts as a magnet for talented individuals and creates an entrepreneurial organizational culture. Garnett's (1992) discussion of public sector communication strategies explains how morale, commitment to agency performance, and the appropriate use of discretion to accomplish objectives, can be engendered among personnel by communicating messages about agency mission and performance rather than overemphasizing procedural issues.

Sustainability transitions has a clear, socially oriented end-goal to shift economic paradigms (Freeman, 1992) or technological regimes (Kemp, 1994) to achieve decarbonization.[11] Sustainability transitions presents guidance on how to recognize good or bad innovations, and might conflict with innovation policies geared toward enhancing productivity and economic growth (Alkemade et al., 2011). Also, accomplishing the sustainability transitions mission will require new policy models. Mowery et al. (2010) emphasize the danger of previous mission-oriented models like the Manhattan and Apollo projects. These projects involved small groups searching for a technology breakthrough, with government acting as the sole buyer. In contrast, a sustainability transition involves multiple technologies, and multiple actors acting as both producers and users. Thus, while the high-level mission of decarbonization might provide guidance, it must act as a motivator for engaging in a process of learning and experimentation rather than accomplishing a single technological feat.

4.5. Embeddedness within policy networks

Technology-specific policy pitfalls could occur because the state cannot acquire sufficient information to make good decisions about which technologies to support and how to support them. Both private and public sectors manage information deficits by creating institutions to acquire, absorb and share knowledge. In his seminal study on the computer industry, Evans (1995) showed how state actors can increase their understanding of the sectors they aim to influence through consistent, sustained linkages with private actors in those sectors. He described the "embedded" relationship the state had with private sector organizations. Embeddedness creates a better understanding of business strategies and likely responses of private actors to policy changes. It builds trust, which facilitates negotiation and coordination on how to achieve policy goals, and increases the likelihood that private actors will share relevant information with state representatives.

Evans (1995) relied heavily on the Korean example, where centralized and coordinated business groups worked in concert with a lead agency. However, he also noted the potential for business to withdraw political support for state involvement after becoming internationally competitive. Thus, Evans (1995) argued that embeddedness must be expanded beyond closed business-state relationships towards incorporating civil society, to maintain the state's capability to influence economic trajectories in the public interest (see also Evans, 2008).

Implementing technological innovation systems policies requires embeddedness. Innovation system actors are asked to uncover system strengths and weaknesses (see Hekkert et al., 2011), and the development of technology-specific advocacy coalitions plays a critical role in building political legitimacy for policy interventions (Jacobsson and Bergek, 2004). We can also understand the challenges created by embeddedness through the multi-level perspective (Geels, 2002). Consider situations where actors supporting technologies that move from niche to regime later seek to reinforce the regime by encouraging innovation through market mechanisms rather than activist government policies. This is akin to the breakdown in transformative business-government alliances described by Evans (1995).

The Netherlands Transition Project is an example of promoting embeddedness in the private sector and civil society. Kern (2011) describes an "open culture" in the civil service that facilitated collaborations with non-governmental actors. van der Loo and Loorbach (2012, p. 240) discuss how such policy networks sought to "strike a balance between engaging regime-

[11] The concept of sustainable development incorporating environmental, social and economic objectives is less clear, but it is a useful guide for social learning and political negotiation (Meadowcroft, 2005).

actors and facilitating front-runners." They also describe the Interdepartmental Program Directorate leading the transition project as an embedded organization that picked up ideas from non-governmental policy networks and introduced them into government (van der Loo and Loorbach, 2012, p. 235).

4.6. Autonomy from private interests

There is a danger that government representatives or civil servants become *captured* by groups seeking to further their private interests by exploiting the close relationship between governmental and non-governmental actors that embeddedness requires. This is why Evans (1995) called for state autonomy alongside embeddedness. Such "embedded autonomy" would allow the state to acquire information and competence while avoiding problems associated with capture. Evans argues that the state must engage with the private sector, but on its own terms.

The state requires a degree of autonomy from private interests if it is to play the unique role in redirecting and shaping technological trajectories emphasized by authors like Mazzucato (2013). As Savoie (2015) argues, we rely on the state to think long-term and solve wicked problems, because other actors lack the means or willingness to do so.

Evans (1995) proposed the principle of autonomy to avoid the worst forms of capture, like corruption or the creation of rents for industrial actors with no public purpose (see Schmitz et al., 2015). The notion of the regime in sustainability transitions studies, however, highlights the more pervasive and subtle issue of incumbent actors influencing technological paradigms and governance systems. Critical review of the Netherlands' transition project noted how the strong influence of incumbent actors led to reduced emphasis on social-institutional reforms and implementation of policies applying transformational pressure on incumbent actors (Kern and Smith, 2008).

Sustainability transitions demands a kind of autonomy the permits state actors sufficient room to envision and explore unique socio-technical trajectories. This must work in conjunction with a form of embeddedness that welcomes non-incumbent and civil society influence through the creation of diverse policy networks. van der Loo and Loorbach's (2012) reflections on the Netherlands transition project resemble a call for embedded autonomy. They note the state's capacity to act as a catalyst for transitions, while recognizing the need for diverse perspectives to pull the state away from acting as a member of the fossil fuel energy regime.

4.7. Competence

Competence is required to counter concerns that the state lacks sufficient market knowledge to make intelligent policy interventions. This lack of information might not only result in ineffective policy, but could contribute to political capture, as the state becomes over-reliant on information from self-interested industry groups. One way to avoid these problems is to ensure the state develops its own competencies and expertise. The development of competence can support both principles of autonomy from private interests and embeddedness, since private actors will engage with the state to access its information and expertise.

Technology-specific knowledge is required to effectively implement technology-specific policy supports. Jacobsson and Bergek (2011) call for deep "domain specific competencies." They suggest that such expertise prevents policymakers from succumbing to technology fads and hypes (see Verbong et al., 2008). Narayanamurti et al. (2009) call for leaders with scientific/technical and managerial credentials, able to understand an organization's role within its innovation system and to "integrate the different activities within and outside the institution." In sustainability transitions, public servants might need to not only focus on a particular sector or technology, but also consider inter-sectoral linkages. Cooke (2012a) argues that sustainability transitions from a regional perspective requires strategies that pursue knowledge interactions across sectoral boundaries. He discusses how Danish wind energy evolved from the agricultural sector, and the transition of a Swedish pulp-and-paper mill toward bio-refining (Cooke, 2012b). Cooke favours creating policy platforms that can "identify and connect" expertise (Cooke, 2007, p. 191), noting the need to develop organizational competencies that understand specific sectors and the advantages of wider technological combinations. The potential for inter-sectoral linkages demonstrates how competence can also complement comprehensiveness.

Lipsey and Carlaw (1998) and Rodrik (2007) emphasize organizational competency. They argue that policy approaches that fall short of theoretical ideals should be prioritized if they are more likely to succeed given pre-existing organizational expertise. However, some warn against overloading organizations deemed to have high capabilities. Breznitz and Ornston (2013) chronicle how institutional capabilities of "peripheral" innovation agencies decline as they attract more political attention. Furthermore, organizations are less likely to prioritize new missions that are imposed on them if they conflict with their primary mission (Frederickson et al., 2016, p. 55). Thus, seeking competence may inadvertently hinder principles of mission-orientation and autonomy.

4.8. Credibility

Ambiguous and politically charged policy environments can produce uncertainty and incoherence. The ability to change policies is a basic requirement of any functioning democracy. However, too much volatility can significantly impede the success of technology-specific policies. Removing risk for private entrepreneurs and/or enticing private firms to enter new sectors or technological areas are primary aims of policies promoting economic transitions (Evans, 1995; Mazzucato, 2015).

Governments must be able to issue signals that private firms can trust (Johnson et al., 2013). The need for credibility can relate both to supply-side incentives that might reduce financing risk, and to government's role as a buyer and a shaper of market demand through mechanisms like regulations and carbon pricing.

Credibility relates to the "influence on the direction of the search" function within the technological innovation system framework (Bergek et al., 2008). Despite the existence of beneficial policy instruments, the perception that such supports might not be sustainable can significantly discourage entrepreneurs or increase the costs of technology deployment (Meijer et al., 2007, 2010; Holburn et al., 2010).

There is a tension between credibility and flexibility principles. Yet, maintaining credibility with the private sector need not imply policy stasis, as this could create a situation that resembles political capture. However, it is reasonable for the private sector to expect predictable policy change, through regularly scheduled reviews (Holburn, 2012) or via policy designs, like price digression schedules for feed-in tariffs, which balance flexibility and certainty (del Río, 2012).

4.9. Stability

The wicked problem of transitioning to a sustainable economy is characterized by uncertain solutions and the need for long-term policy efforts. Long-term objectives may coincide uneasily with government funding cycles and inconsistent political prioritizations. Foray et al. (2012, p. 1701) call for stability in an organization's "broad orientation and funding," and Narayanamurti et al. (2009) call for "stable, sufficient, and predictable" funding for projects that might take several years. A boom-bust funding cycle can decrease organizational competency and lead to information problems. Rapid funding increases might not coincide with the longer process of building human capital, and funding decreases can lead to the stranding of the skills developed within a project (see Newell, 2008; Freeman and Van Reenen, 2009).

The transitions management approach includes long-term visions to guide policy processes (Rotmans et al., 2001), which could support stability. However, the Netherlands example demonstrates the difficulty of maintaining policy continuity. In 2011, the energy transitions project was reduced in scope by a new government implementing budget cuts and facing international competitiveness concerns (Laes et al., 2014).

4.10. Accountability

Principal-agent problems can arise from information asymmetries between elected representatives and civil servants. Doctrines like "ministerial responsibility," in the Westminster system of government, make elected representatives ultimately accountable for all civil service decisions. Accountability to elected representatives and the public means that government organizations must be transparent and obliged to "explain what they are doing and how they are doing it" (Rodrik, 2014).

Information on activities and performance can be conveyed through evaluation. Sustainability transitions frameworks emphasize the need to track multiple indicators. The technological innovation system approach tracks various innovation activities and processes to uncover socio-technical barriers (Suurs and Hekkert, 2012). The multi-level perspective also provides a multi-faceted evaluation of socio-technical transitions (Kern, 2012). Yet, communicating results is complicated by the complex, evolutionary nature of transitions, where cause-effect relationships might be uncertain. The Netherlands transitions project included monitoring and evaluation procedures that aimed to clarify how the transition was unfolding, the threats and opportunities of current pathways, and consideration of relevant changes to experiments, long-term visions and pathways. However, such evaluation required significant organizational capacity and time commitments, and participants found certain transition patterns used to inform the process too abstract (Taanman, 2012).

Interpreting evaluation results is complicated by different ideological perspectives on objectives and preferred policy instruments, and thus accountability involves engaging in democratic discussion and debate. Without strong political consensus, democracies produce overlapping, competing objectives, and might have different perceptions of good or bad performance. This can increase policy ambiguity and reduce mission orientation. Organizations that aim to be accountable and effective, must strive to communicate benefits and gain legitimacy with the public and elected representatives.

5. Managing the principles in practice

Table 1 presents a synopsis of the discussion in the previous section of how the institutional design principles discussed alleviate potential policy pitfalls. The dots indicate the policy pitfalls an institutional design principle helps alleviate or avoid. The policy pitfalls are listed under the four theoretical categories discussed in Section 2: information failure; political capture; principal-agent problems; and complexity, ambiguity & risk aversion. The table shows that the principles present multiple, overlapping, solutions to each policy challenge.

While, Table 1 signals that the principles help solve potential policy problems, the table does not represent either the challenges associated with realizing the principles, or the tensions and contradictions that may occur between principles. For instance, these principles call for ensuring both accountability and autonomy from political pressure; for flexibility and a stable policy approach that builds credibility; for embeddedness within the private sector, yet autonomy from private interests. Working with tensions is a regular part of public management, and thus how these principles operate in practice

Table 1
The Policy Pitfalls Alleviated by Institutional Design Principles.

Institutional design principles	Policy pitfalls			
	INFORMATION FAILURE	POLITICAL CAPTURE	PRINCIPAL-AGENT PROBLEMS	COMPLEXITY, AMBIGUITY & RISK AVERSION
Comprehensiveness	●			●
Flexibility	●		●	●
Autonomy from Short-term political pressure				●
Mission-orientation			●	●
Embeddedness within policy networks	●			
Autonomy from private interests		●		
Competence	●	●		
Credibility				●
Stability	●			●
Accountability			●	

and how they interact with one another is best explored through a case study. Thus, this section explores the institutional design principles through the case of the Advanced Research Projects Agency-Energy (ARPA-E) in the United States.

5.1. Introduction to ARPA-E

ARPA-E is an agency within the United States Department of Energy that promotes high-potential, high-impact energy innovations. A Committee on Prospering in the Global Economy of the 21st Century (2007) called for an energy agency modelled on the Defense Advanced Research Projects Agency (DARPA) in order to stem erosion of the US's advantage in science and technology.[12] ARPA-E was created in 2007, but not funded until 2009.

ARPA-E has a small staff (56 full-time equivalents) (ARPA-E, 2016a) and a multi-faceted mission to improve US economic competitiveness, energy security and reduce greenhouse gas (GHG) emissions. It funds short-term projects (1–3 years) with potential for transformative impact. Projects are cost-shared with funding recipients, which have included universities, businesses, national labs and non-profit organizations. ARPA-E is an actor in a larger network of energy innovation institutions incorporating federal labs, innovation hubs and other organizations within the Department of Energy (see Anadón, 2012).

Chu (2006) explained before Congress that a separate, mission-oriented agency would aim to "bring a freshness and excitement to energy research that will attract many of our best and brightest minds" (an example of the mission attracting talent). The organization is given the flexibility to work outside regular government hiring and contracting procedures, enabling it to hire competent staff quickly (Bonvillian, 2014). Program directors serve three-to-five year terms, which creates a sense of urgency and mission (Majumdar, 2011; Martin, 2015).

Here follows a discussion of how ARPA-E's institutions (its norms and rules) have enabled it to create synergies between the principles outlined above and certain principles that ARPA-E struggles to realize.

5.2. Balancing accountability and political autonomy to achieve comprehensiveness

One possible tension between principles outlined in Section 4 is the need to balance political autonomy with public accountability, while using all tools of government to provide comprehensive policy support. ARPA-E takes a comprehensive view of the innovation environment. In evaluating projects, program directors consider how innovations will link with other competing or complementary technologies; market and regulatory risk; and political, cultural and economic risks (Sagar and Majumdar, 2014). In addition, a team of tech-to-market advisors uses its networks and market knowledge to connect technology developers with private sector funders and business partners (Martin, 2015). ARPA-E supported projects are required to spend at least 5% of their funding on activities to facilitate market entry, like techno-economic analyses, market research, IP protection, and engagement with potential customers or investors (ARPA-E, 2016a, p. 399). Rather than follow a "linear model" (Bush, 1945) in the belief that basic and applied research naturally leads toward commercialization, the agency works "right-left" across the innovation pipeline, deciding what kinds of technological breakthroughs must be achieved, and then pushing the boundaries of basic and applied science to meet these objectives (Bonvillian, 2014). As Martin (2015, p. 108) explains, ARPA-E regularly asks, "if it works, will it matter?"

ARPA-E promotes technologies developed for concrete end-uses, but its core function within the innovation system is not to promote scale-up and diffusion. Rather, it promotes experimentation and searches for novel solutions to diversify the technological options available (Martin, 2015, p. 106). The agency operates as an arm's-length organization, giving it room to experiment with new technologies, while avoiding the risk aversion that can occur in organizations confronted with short-term political pressures. However, this position creates the potential drawback of the agency losing control over

[12] DARPA was created in response to concerns, following the Soviet launch of Sputnik in the 1950s, that US scientific capacity was eroding.

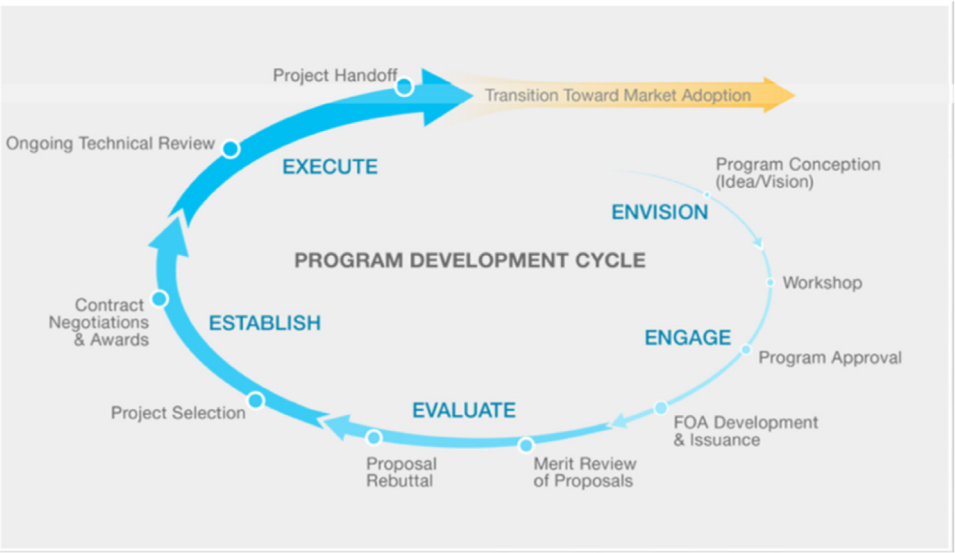

Fig. 1. ARPA-E Program Development Cycle.

Source: ARPA-E, 2013.

its projects as they move toward advanced innovation stages. For instance, the agency lacks the ability to create initial niche markets (readily available to DARPA through the Defence Department's procurement power).[13] Concerns about political accountability may also arise from arm's-length status.

ARPA-E manages these tensions through an "island-bridge" model, as described by Bonvillian (2014). In this conceptual model, ARPA-E has an island where it is free to experiment, but it also has a direct bridge to government through the Secretary of Energy. The bridge creates a direct link with a member of the President's Cabinet to ensure accountability. This link also provides ARPA-E with a high-level political champion capable of working government machinery. The Secretary oversees a number of organizations within the department, and ensures projects are suitably handed off to other agencies focused on deployment and diffusion.[14]

The "island-bridge" model is an example of an institutional arrangement that creates synergies between the principles of autonomy from short-term political pressures, accountability and comprehensiveness. The island provides the political autonomy and the bridge ensures accountability. The combination of both principles helps ARPA-E coordinate with other government agencies to ensure comprehensiveness.

5.3. Achieving embedded autonomy

Section 4 identified embeddedness within policy networks and autonomy from private interests as two institutional design principles that must be realized in tandem. ARPA-E's institutional design supports a mix of embeddedness and autonomy in relation to technology developers. The program creation process runs through five stages: envision, engage, evaluate, establish and execute (Fig. 1). In the "envision" stage, program directors develop a vision of the energy landscape. This leads to program concepts, which are debated through workshops with both internal and external experts. Early feedback is sought from the business community and other technology developers, but it is important to highlight that this stage is meant to make program developers "define success early before others define it for you" (Sagar and Majumdar, 2014). This ensures program directors have the requisite autonomy to make independent judgments.

Other elements of the program development process ensure some degree of embeddedness and information exchange with those outside of the organization. ARPA-E seeks to engage diverse communities, including regulators, suppliers, customers and those not traditionally involved in the project area (Martin, 2015). External reviewers evaluate program proposals, and applicants are invited to provide rebuttals. In the "engage" stage, program directors encourage competition for ideas between participants, which funnels information to the program directors and avoids capture by one group by inviting a diversity of perspectives.

Embedded autonomy is further achieved through ARPA-E's open solicitations for proposals and focused challenges in particular technological areas. The focused challenges follow ARPA-E's vision of promising technological trajectories, while

[13] On DARPA and military procurement, see Block (2011).

[14] ARPA-E also shares a memorandum of understanding with the Department of Defense to test, validate and offer initial markets for its technologies (Bonvillian, 2014; Mabus, 2011).

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the open solicitations invite idea flows from outside the organization. In 2013, ARPA-E launched an IDEAS program to provide small, short-term grants for early-stage, unproven technical concepts that could become focused technology programs. These latter two initiatives attract solutions that ARPA-E program managers might not have considered.

A mix of embeddedness and autonomy also arises in ARPA-E's selection of program directors, who are not permanent employees, but individuals from academia and industry who serve three-to-four-year terms. Thus, program directors bring knowledge gleaned over careers embedded in energy research and/or entrepreneurship. However, while at ARPA-E, the program directors represent the organization, not industrial or academic interests, which maintains autonomy.

5.4. Cooperative agreements: flexibility enhanced through competence, embeddedness and credibility

Flexibility enables an organization to change strategic approaches in light of new information and to cut off non-performing projects in sufficient time. This principle is not easily achieved, yet ARPA-E demonstrates that flexibility can be supported by finding synergies with other principles.

ARPA-E's cooperative agreements signed with funding recipients are unlike a standard grant. The main difference is that program directors have "substantial involvement," which means they do not limit themselves to administration, but may also request changes to project teams and redirect technical aspects (ARPA-E, 2012). This is an example of a highly embedded relationship.

While program directors have the contractual right to become substantially involved, their deep technical understanding ensures their role in the development team. Thus, competence and embeddedness reinforce one another. This combination of embeddedness, which gives program directors access to relevant information, and competence to exercise sound judgments with that information, equips ARPA-E with the flexibility to modify or cancel projects.

It is also important to consider the funding recipients' motivations. They might become non-cooperative and seek to hide information to avoid program cancellation. To prevent this, ARPA-E creates a culture where there is no shame in failure (Biello, 2011) and fosters the understanding that "ideas fail, but people don't" (Sagar and Majumdar, 2014). This norm assures participants that engagement with a failed project will not count against them. It is also critical that technology developers see ARPA-E's commitment not to penalize failure as *credible.* The maxim "ideas fail, but people don't" can steer technology developers away from defending non-performing technologies. It can promote trust and mutual learning between ARPA-E and funding recipients, further reinforcing the information exchange benefits of embeddedness and creating greater flexibility to cancel or modify projects.

5.5. Searching for stability & the politics of accountability

ARPA-E is accountable to the President as well as Congress, and must contend with the different political ideologies and priorities of the branches of government in annual budget negotiations. The organization struggles to maintain budget stability and to garner enough resources to deliver on its original mission. Its stability is also vulnerable to changes in political leadership.

The Committee on Prospering in the Global Economy of the 21st Century (2007) recommending the creation of ARPA-E foresaw an annual budget level that gradually increased to $1 billion in five to six years. $403.8 million in initial funding for ARPA-E came through the American Recovery and Reinvestment Act of 2009, with another small amount through Congressional appropriations. The call for economic stimulus in the wake of the recession created a political window that provided the initial funding. Starting in fiscal year 2011, ARPA-E presented annual budget requests through the President to Congress. Fig. 2 presents budget requests and the actual funding levels following negotiations. Annual budgets remained relatively steady at the $200-$300 million range. However, this steady outcome masks the volatility of the budget setting process, which is characterized by several proposals and counter-proposals. For instance, the Energy and Water Development House of Representatives appropriations subcommittee recommended a 44% cut from the previous year's funding in 2012, and a 72% cut in 2014 (Behrens, 2012, 2013). During these proceedings, some have warned that funding instabilities could reduce credibility with private sector partners and compromise ARPA-E's ability to attract high-quality personnel (influencing competence) (Anadón, 2012; Hourihan and Stepp, 2011).

The annual budget process demonstrates how ARPA-E must be accountable to the Secretary of Energy, Congress and the institutions responsible for evaluation. Audit reports scrutinized eighteen (of 121) cases where ARPA-E projects had previously received private sector investment (Government Accountability Office, 2012), and the appropriateness of $40,000 in technology transfer and outreach costs (Department of Energy, 2011). In response, one Congressional subcommittee chairman proposed restricting ARPA-E's ability to award funds to projects based on technology readiness level (see Broun, 2012). This proposal demonstrates the difficulties ARPA-E faces in ensuring flexibility and comprehensiveness when faced with ideological debates over the role of early-stage R&D versus a technology-specific approach that considers barriers across all stages of development. Subsequent analysis of the questionable technology transfer and outreach activities found that 98% of the costs were allowable, and ARPA-E followed Audit Office recommendations to provide increased information explaining private investors' unwillingness to fund ARPA-E supported activities (Majumdar, 2012).

To improve stability, ARPA-E has attempted to remove some of its funding from the annual budget cycle. The first attempt was a 2012 request for $100 M in mandatory funding, not contingent on annual appropriation bills from Congress. The funds would have used wireless technologies to improve electrical infrastructure, energy efficiency, and transportation systems

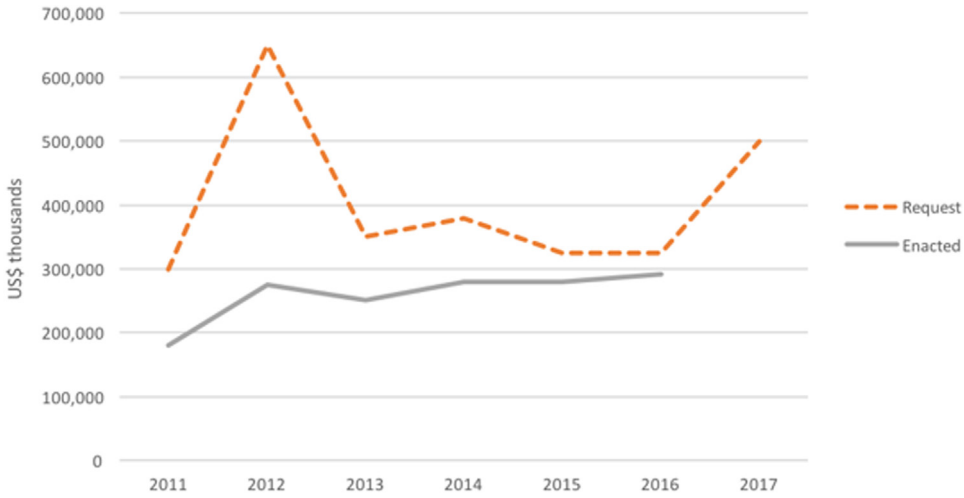

Fig. 2. ARPA-E Funding.

Source: ARPA-E (2016b).

(see Majumdar, 2011). For fiscal year 2017, the President proposes to create an ARPA-E Energy Trust of $1.85 million over five years. The agency notes the need for a "reliable stream of funding" (ARPA-E, 2016a, p. 397). The extra funding would enable ARPA-E to take on "larger scale, and more complex projects" and "systems level" efforts aimed at integrating multiple technologies and making large impacts on energy systems. For instance, ARPA-E notes the potential to explore coupling projects focused on thermal energy management, heat-to-electricity and solar electricity (ARPA-E, 2016a, p. 420). ARPA-E also warns that some projects could be "orphaned," as further time is required to create prototypes and to demonstrate market readiness through field testing and scale-up assessment (ARPA-E, 2016a). The agency has requested a stable and sufficient budget so it may develop competencies in exploring cross-technology interactions and provide comprehensive policy supports, which include preparing technologies for market entry.

The Senate Appropriations Committee chairman has stated that the 2017 budget request contains "out-of-control mandatory spending" (Alexander, 2016), and the House of Representatives and Senate both approved appropriations lower than the President's discretionary budget request.[15] Despite the difficulties in approving mandatory spending and budget increases, ARPA-E receives relatively strong bi-partisan support compared with other energy programs. Republicans have criticized the President's larger "Mission Innovation" proposal to dramatically increase energy R&D for favouring the Office of Energy Efficiency and Renewable Energy over basic R&D, fossil fuel and nuclear technologies (Ambrose, 2016).

ARPA-E faces stability challenges because a broad political consensus on climate change does not exist in the United States (Anadón, 2012, p. 1753).[16] The 2016 election of President Trump, who called climate change a hoax and promised to increase fossil fuel production, threatens ARPA-E's existence and the Department of Energy's current agenda. Interviews with department staff found that the shift in mission could result in a mass exodus of talent and leave energy agencies vulnerable "to attack in a hyper-partisan era" (Lacey, 2016). ARPA-E might need to rely on the bi-partisan support it has achieved in Congress as it loses the President's Office as its chief advocate.

Stability is ARPA-E's primary challenge. The agency searches for stability to reinforce policy comprehensiveness and to fulfill its mission. Stability exists in tension with the need to be accountable to multiple elected representatives when a consensus on climate change and the role of technology-specific policies does not exist. As ARPA-E requests more funds it must open itself up to greater democratic accountability, which could introduce scrutiny that limits flexibility and subjects it to short-term political pressures. The agency must now contend with a major change in political leadership, which could dramatically alter its mission, reduce its funding, or even threaten the organization's survival.

6. Conclusion

Policymakers seeking to implement more technology-specific approaches to spur environmental innovation and sustainability transitions must explicitly address the issue of institutional design. Pitfalls created by information deficiencies, political capture, asymmetries between elected principals and implementation agents, and risk aversion stemming from

[15] At the time of writing the final agreement on fiscal year 2017 spending has yet to be achieved. The "lame duck" Congress passed a continuing resolution to fund the government. The new Congress and President, following the 2016 election, now have the responsibility of negotiating and approving the 2017 budget.

[16] Block (2008) discusses how the political consensus supporting defense R&D and procurement has enabled the implementation of US industrial policy.

policy ambiguity and political blame could make implementation of technology-specific policies less effective. Recognition of these pitfalls necessitates closer analysis of principles and practices that allow the public sector to work effectively.

This article describes and discusses ten institutional design principles to promote the effective implementation of technology-specific policies: 1) comprehensiveness; 2) flexibility; 3) autonomy from short-term political pressure; 4) mission-orientation; 5) embeddedness within policy networks; 6) autonomy from private interests; 7) competence; 8) credibility; 9) stability; and 10) accountability. These principles can alleviate policy pitfalls, yet can be a challenge to realize. Many of them pull policymakers in different directions and sometimes appear contradictory.

The ARPA-E case demonstrates how these principles can become mutually reinforcing and produce creative tensions. The "island-bridge" structure seeks to combine the benefits of accountability and autonomy from short-term political pressure, enabling more comprehensive policy supports by integrating ARPA-E into the larger network of energy innovation organizations. ARPA-E can also achieve the seemingly contradictory principles of autonomy from private interests and embeddedness within policy networks. The agency's program development cycle sees directors developing their own visions of technology futures, which supports autonomy in engagements with outside parties and helps avoid political capture. ARPA-E also achieves flexibility to terminate and re-scope projects by ensuring program managers have "substantial involvement" (embeddedness) and competence, while the informal norm "ideas fails, but people don't" builds credibility and trust.

The ARPA-E case also shows that the tension between stability and accountability is difficult to manage. Accountable to both the President and Congress, ARPA-E must navigate annual budget negotiations between the different branches of government. Attempts to increase resources and funding certainty have been unsuccessful. Increased funding would align ARPA-E with its original mission to reclaim US science and technology leadership, and funding stability would enable support for long-term and system-level projects. The difficulty of developing a stronger political consensus behind sustainable energy and technology-specific policies creates instabilities that limit or threaten to compromise several principles, including mission-orientation, comprehensiveness, competence, political autonomy and flexibility. The results of the 2016 election could dramatically change ARPA-E's effectiveness and orientation.

This paper proposes ten institutional design principles that can address technology-specific policy pitfalls. The ARPA-E case describes how many of these principles are achieved through institutional and organizational innovation, and it presents an example of tension between principles. Further case study research is required to explore these principles in different contexts and the means by which effective institutional design is achieved. The ability of public sector institutions to implement technology-specific policies will depend on certain contextual conditions[17]: pre-existing competencies in the public sector, the characteristics of larger policy networks, and political environments. The principles and case study example herein do not comprise a magic policy prescription. Rather, they are meant to inform the context-specific search for public sector institutional configurations that work.

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[17] Bergek et al. (2015) describe how technology innovation system strengths and weaknesses are influenced by different contexts. Policy implementation will also be influenced by context conditions.

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